

EDUCATION

University of Michigan Ann Arbor | *M.S. Applied Data Science*
University of California Santa Barbara | *B.S. Computer Engineering*

Aug 2025 - Present
Jun 2024

WORK EXPERIENCE

Applied Materials | *Software Engineer* [AIx](#) | Python, Machine Learning Santa Clara, CA | July 2024 - Present

- Creating a data search platform to accelerate time to market for semiconductor process development.
- This platform aggregates sensor data, automatically links experiments done by process engineers to metrology results, and allows for searching or generation of new processes to accelerate R&D.
- Serving as a core developer from initial proof-of-concept (Aug 2024) to production (Feb 2025), now achieving 75% adoption in Etch division (600+ monthly users).

KLA | *Machine Learning Engineer Intern* - [BBP](#) | Python, Image Classification Milpitas, CA | June-Sept 2023

- Developed a machine-learning autoencoder using TensorFlow for BBP Nuisance Filtering, reducing cluster count by 30% in IBM CAD and Samsung GMK dataset compared to DBG2 (current implementation).
- Engineered an advanced, adaptable framework using convolutional layers, optimizing defect detection and classification across multiple datasets, specialized by training with unlabeled BBP hotspot images.
- Utilized HDBSCAN for efficient clustering of latent features, validated by silhouette scores.

Synopsys | *Application Development Intern* | Java, SQL, Data Processing Santa Clara, CA | June-Sept 2022

- Conducted performance analysis and optimized algorithms for a branching process flow resulting in a reduction of server load processing time from 10 minutes to 2.5 minutes.
- Automated contractor onboarding process saving 2,200 employee hours annually, improving system efficiency for over 2,000 users.

PROJECTS

[Pixel Pane – AI Assistant](#) | SwiftUI, MLX, Cloudflare Workers, Anthropic API May 2026

- Built model-agnostic SwiftUI macOS AI assistant supporting any MLX-compatible backend, Anthropic Claude, and Apple Foundation Models — with per-provider health detection and lifecycle management.
- Deployed Cloudflare Workers proxy normalizing Anthropic streaming into SSE with per-user rate limiting, KV quota enforcement, and cryptographic request tracing.
- Engineered extensible action dispatch system with heuristic content classification, built-in handlers for translation and code debugging, and user-defined side effects.
- Designed permission-gated security model with explicit file grants and scoped side-effect authorization before any system-level operation.

[iso-code – Safe Worktree Management for AI Agents](#) | Rust, CLI, MCP Server, Git Apr 2026

- Built a Rust library, CLI, and MCP server providing safe git worktree lifecycle management for AI coding agents across Claude Code, Cursor, and Copilot.
- Prevented data loss from unmerged-commit deletion, orphaned worktrees, and git-crypt corruption with atomic state writes and cleanup-on-failure guarantees.

[Health Monitoring Wearable – Capstone](#) | Python, C, Java, ASIC Design Nov 2023-June 2024

- Designed and manufactured a specialized health monitoring wearable for nursing home residents, focusing on comprehensive remote monitoring and enhanced data accuracy.
- Incorporated advanced sensors in the wearable for vital signs, activity tracking, temperature monitoring, ambient noise detection, and continuous heart rate and oxygen level measurement.
- Engineered a data-efficient Android app for real-time health data visualization and trend analysis.
- Designed and manufactured a 4-Layer PCB design in Altium.

SKILLS

Languages: Python, SQL, C++, C

ML & Data: PyTorch, TensorFlow, ClickHouse, DuckDB

Infrastructure: Kubernetes, Kafka, AWS (MSK, S3, CloudWatch), Docker, FastAPI, Git

AI tooling: MCP, MLX

Neural Networks and Deep Learning | [deeplearning.ai](#)

Jan 2022